Solution Requirements

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-1660
Project Name	AI WEATHERWISE API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/ Low)
1	Authentication	Users can register, log in, and securely access the system using JET-based authentication.Passwords are Securely stored using encryption /hashing.	High
2	Authorization levels	The system provides role-based access for **Admin, Editor, Author, and Reader**. Each role has specific permissions for managing blogs and content.	High
3	External interfaces	The API connects with **Google Gemini AI** for AI-powered blog generation and assistance.	High

		It also connects with **MongoDB** for storing application data.	
4	Transactions processing	The system handles user requests such as creating, updating, viewing, and deleting blogs, categories, tags, and comments while maintaining data consistency.	High
5	Reporting	The system provides blog and content information through API responses, including blog details, categories, tags, comments, and user-related information	Medium
6	Business rules, etc.	Users can create and manage blogs according to their assigned roles. Admins can manage users and content, while authors/editors can manage permitted blog content.	High
7	Compliance to laws or regulations	The system should protect user credentials and personal information through secure authentication, password hashing, authorization, input validation, and secure API practices.	High
8	*Other*	The system supports AI-assisted content generation, FAQ assistance, content search, categories, tags, comments, error handling, and API validation.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The API should process user requests and return responses quickly. AI-generated content should also be processed efficiently.	API **response time < 2 seconds** for normal requests; AI responses should be generated within a reasonable time.
2	Scalability	The system should support an increasing number of users, blogs, and API requests without major performance issues.	Should support **multiple concurrent users** and allow database growth without significant performance degradation.
3	Security & Data Privacy	User passwords and personal information must be protected. Authentication, authorization,input validation, and secure API communication should be implemented.	**JWT authentication, password hashing, input validation, and HTTPS**; unauthorized users must not access protected resources**.**
4	Reliability & Availability	The API should operate consistently and handle errors without crashing the entire system.	Target **99% availability**; errors should return proper HTTP status codes and meaningful messages.
5	Usability & Accessibility	The API and its connected application should be simple, understandable, and easy for users to interact with.	Clear API responses, simple navigation, readable messages, and responsive interface on **mobile and desktop**.
6	*Other*	*Add any additional non-functional requirements*	*Maintainability, Portability*